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KEYNOTE: “BILLINGHAM CUTTING EDGE LECTURE” – THE HISTORY OF THE IAA SETI PERMANENT COMMITTEE – 1990 TO 1999

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Abstract

This paper is the third in a series exploring the history of the International Academy of Astronautics (IAA) SETI (Search for Extraterrestrial Intelligence) Permanent Committee. Established in 1974, the committee was formed in response to growing awareness that humanity might detect evidence of intelligent life beyond Earth. The IAA SETI Committee has an exceptional mandate to examine all aspects of the search for extraterrestrial intelligence. This encompasses a range of topics, including international collaboration, astrophysical observations, biochemistry, exoplanetary research, evolutionary biology, space missions, search methodologies, and the societal, legal, and political ramifications of a confirmed detection. The IAA SETI Committee is the only international entity that meets consistently yearly to organize symposia, select papers for publication, address specialized topics in SETI, and collaborate with other organizations. This paper focuses on the committee's activities throughout the 1990s – a pivotal and turbulent decade for the global SETI community. Between 1990 and 1999, the IAA SETI Committee continued its vital work in disseminating the “Declaration of Principles Concerning Activities Following the Detection of Extraterrestrial Intelligence”. This seminal document was approved in 1989 by the Board of Trustees of the IAA, the Board of Directors of the International Institute of Space Law, and other organizations. The contentious issue of how to respond to a verified detection of extraterrestrial intelligent life or even whether communication as such should be instigated was dominant throughout the 1990s as the IAA SETI Committee worked on drafting an IAA Position Paper on the matter. By the end of the 1990s, public interest in the scientific search for extraterrestrial intelligent life in the universe had amplified due to the astounding verification of a multitude of extrasolar planets around other stars, the announcement of potential fossil life on Mars, popularity of the critically acclaimed movie *Contact*, false claims of detection of radio signals from extraterrestrial intelligence, and the rocketing interest in the groundbreaking SETI@home program. The IAA remained steadfast in its support and promotion of the committee's activities and in seeking United Nations recognition of the consequences to humanity of detecting extraterrestrial intelligent life in the universe.

Keywords: SETI, IAA, Protocol, Transmissions, Declaration

1. Introduction

A paper describing the founding of and early years of the IAA SETI Committee* was presented at the 66th IAC in Jerusalem, Israel, in 2015 [1]. A second paper covering activities of the IAA SETI Committee and events pertaining to global SETI efforts during the period 1980 to 1989 was presented at the 70th IAC in Washington, D.C. in 2019 [2]. This paper chronicles the activities and achievements of the IAA SETI Committee during the decade spanning 1990 to 1999.

1.1 The International Academy of Astronautics (IAA)

After the Second World War, Germany, Britain, and France organized an international meeting for members of rocket societies and other groups and individuals to

discuss non-military space research, space travel, and related topics. The first International Astronautical Congress (IAC) was held in Paris in 1950 and the International Astronautical Federation (IAF) was subsequently established to foster dialogue between scientists and support international cooperation for the non-military use of space, including the development of astronautics and interplanetary flights.

In 1957 humanity entered the space age, with the launch of Sputnik. In 1960 the IAF, an organization of member-societies, approved the formation of an international organization of distinguished individuals in the aeronautics field, the International Academy of Astronautics (IAA). The International Institute of Space Law (IISL) was established at the same time. The IAA provides a non-governmental forum through which members can contribute to advances in all aspects of the peaceful exploration of space, including space policy, law, economics, and space and society. The IAA and the IISL would, under the umbrella of the IAF, hold joint

* In this paper, the IAA SETI Committee refers to the International Academy of Astronautics SETI Permanent Committee. The IAA SETI Committee was known as the IAA CETI Committee until 1987.

symposia and meetings at the annual International Astronautical Congresses. This structure remains in place today.

1.2 Introduction of SETI to the IAA

When the IAA was established, the space age was underway, and there was a growing realization that humans might detect extraterrestrial life, including intelligent life.

Project Ozma, the first modern scientific search for radio signals from extraterrestrial intelligence took place at Green Bank, West Virginia, U.S.A. in 1960. Rudolf Pesek, a member of the new IAA, valiantly introduced the topic of SETI to his IAA colleagues in 1965. In 1972 the first IAA CETI symposium took place during the 23rd IAC in Vienna, Austria. By 1974 the IAA established the SETI Committee with Rudolf Pesek appointed as Chair. Of particular importance was the view of the new committee that papers, talks, and activities should address both the technological and sociological aspects in the search for intelligent extraterrestrial life in the universe.

1.3 IAA SETI Committee Mandate

The mandate of the IAA SETI Committee is as follows:

1. Examine all aspects of possible future contact with extraterrestrial civilizations, with special reference to international issues and activities involving the underlying science of the study of life in the universe including:
 - relevant astrophysical and astronomical phenomena
 - the nature and distribution of biogenic elements and compounds
 - the formation of life-bearing planets
 - the origin and evolution of biological systems
 - complex life and intelligence
 - the emergence of technological civilizations
2. The SETI Committee also looks at associated space missions and technology including:
 - planetary science missions designed to search for evidence of extraterrestrial life
 - the astronomical search for extrasolar planets
 - the recovery of possible Martian microfossils on the Earth
3. The science and technology of SETI includes:
 - the development of search strategies and the underlying intellectual foundation
 - the design, development, construction and operation of telescopes and signal detection

systems, studies of their future location in space or on the lunar farside

- rejection of radio frequency interference
- the archiving of SETI data, and analysis of search results.

And if that was not enough, the IAA also states that the IAA SETI Committee will continue to lead discussions on the implications of detecting extraterrestrial signals. Post-detection areas of study include philosophy, historical analogues, anthropology, legal, political, and institutional issues, sociology, psychology, theology, media, and education.

Finally, the IAA SETI Committee organizes annual SETI sessions/symposia as part of the IAF-sponsored annual International Astronautical Congress (IAC). Traditionally, the first SETI session focuses on technical papers, and the second session is devoted to social sciences and interdisciplinary studies. The committee selects peer-reviewed papers of interest for publication in the scientific journal *Acta Astronautica*, works cooperatively with other groups to co-sponsor conferences, meets annually to discuss SETI matters, and establishes taskforces, workgroups, and subcommittees to grapple with issues of importance to SETI activities. The IAA SETI Committee is to conduct preparations for the detection of intelligent extraterrestrial life and consider the aspects and implications of a potential reply from Earth.

1.4 IAA SETI Committee – Early Activities and Accomplishments

One of the inaugural actions undertaken by the new IAA SETI Committee was to advocate for the protection of particular radio frequencies to facilitate the search for extraterrestrial signals on specific wavelengths of the radio spectrum. The resolution was approved by the IAA and IAF, but ultimately, the International Telecommunications Union (ITU) declined to allocate dedicated space in the spectrum for SETI activities. The IAA SETI Committee also collaborated with the International Astronomical Union (IAU) to sponsor sessions and support Bioastronomy Commission Symposia.

By 1980 Rudolph Pesek stepped down as inaugural Chair of the IAA SETI Committee and John Billingham assumed the Chair role.

As stated in its mandate, the IAA SETI Committee was never just about the science and technology involved in searching for extraterrestrial intelligent life in the universe. The international, historical, societal, political, legal, institutional, educational, and media

aspects of the search for and the consequences of a discovery of extraterrestrial intelligent life are also considered and studied. The IAA SETI Committee recognized that SETI researchers could benefit from specific principles or guidelines about what happens if a candidate signal is identified. The verification of a signal, data dissemination, and the public announcement of such a momentous discovery are factors flowing from the initial observation.

To address the need to establish a framework of sorts for any candidate signals, members of the IAA SETI Committee worked diligently throughout the 1980s to produce a document outlining a series of guidelines and principles for individuals or organizations engaged in carrying out scientific searches for signs of extraterrestrial intelligence. The committee worked closely with several experts, most notably from the IISL, to draft a series of guidelines that SETI researchers, mainly radio astronomers, should consider if candidate signals are detected. Most of this work was undertaken by a subcommittee established by the IAA SETI Committee chaired by member Michael Michaud. By April, 1989 the final seminal document “Declaration of Principles Concerning Activities Following the Detection of Extraterrestrial Intelligence”, or “Declaration” as it is referred to in this paper, was endorsed by both the IAA and IISL. The Declaration has no formal government status and is voluntary; however it was and remains a remarkable achievement [3].[†] A special issue of *Acta Astronautica* features papers pertaining to the Declaration.[4]

The nine principles listed in the Declaration provide guidance on candidate signal verification, informing other observatories for confirmation, and when to alert authorities, including the Secretary General of the United Nations. The Declaration states that the news of any confirmed detection of extraterrestrial intelligent life should be disseminated promptly, openly, and widely through scientific channels and public media. The discoverer should have the privilege of making the first public announcement. Data should be collected and any critical radio frequencies should be protected. The Declaration states that the IAA SETI Committee in coordination with Commission 51 of the International Astronomical Union, will conduct a continuing review of procedures for the detection of extraterrestrial intelligence. The IAA SETI Committee will initiate and maintain a current list of willing experts and individuals with relevant skills to advise on analysis of a verified signal.

[†] The Declaration document was revised in 2010.

Most notably, Principle 8 in the 1989 Declaration states, “No response to a signal or other evidence of extraterrestrial intelligence should be sent until appropriate international consultations have taken place. The procedures for such consultation will be the subject of a separate agreement, declaration or arrangement”. The issue of a deliberate transmission from Earth has and remains contentious since it was brought up in the 1980s. To address this issue the IAA SETI Committee started work during the 1980s on the question of a “Reply from Earth” with a goal of producing a second Declaration document dealing with this issue.

2.0 1990 – Thirty Years of SETI Searches

The 1990s began on a positive and optimistic note for the IAA SETI Committee. The Declaration document was on its way to being endorsed by international scientific organizations. Simultaneously, efforts were underway to formulate the second document addressing Earth’s potential response to a confirmed extraterrestrial signal.

Notably, the first modern radio telescope search for extraterrestrial signals, Project Ozma, took place thirty years ago. By 1990, over 50 documented scientific searches for signs of non-human technology beyond Earth were completed by SETI search investigators using facilities located in the U.S.A., U.S.S.R., Canada, France, Holland, Argentina and Australia and others.. Most of these searches focused on detecting radio signals through targeted or broadband scans. Others aimed to identify extraterrestrial technology through anomalies in the electromagnetic spectrum, potentially indicative of Dyson spheres, stellar engineering, laser signals, or starship propulsion leakage. Meanwhile, NASA’s high-profile SETI project, the High Resolution Microwave Survey was in advanced planning stages.

The 42nd IAC took place in Dresden, Germany, in 1990. A special highlight of the SETI review sessions was the presentation of Frank Drake’s “Reflections on the Modern History of SETI,” inaugural Pesek Lecture, in honour of Rudolph Pesek, the founding force behind the IAA SETI Committee, who had passed away in 1989.

Present at the 17th annual meeting of the IAA SETI Committee were Co-Chairs Ivan Almar and Jill Tarter, along with committee members Jean Heidmann, Mieczyslaw Subotowicz, William Pritchard, Mike Klein, Aldo Cocca, Roger Malina (representing Stuart Bowyer), Vladimir Kopal and Bill Hilton.

While not physically present at the Dresden meeting, John Billingham, the Chair of the IAA SETI

Committee, was actively promoting multidisciplinary studies on the cultural and social implications of the potential discovery of extraterrestrial intelligent life. He initiated a series of NASA workshops on the cultural aspects of SETI beginning in 1991, continuing into 1992, focusing on the societal implications of potential detection [5]. In the broader astronomical community, efforts continued to verify potential evidence of extrasolar planets, a crucial element in the search for signs of extraterrestrial life.

At the Dresden meeting, it was reported that the SETI Committee had strengthened its collaborative work with other scientific organizations, particularly with the International Astronomical Union (IAU). The IAA was a sponsor of the Bioastronomy Commission Symposium held in Val Cenis, France in June, 1990.[‡] At this symposium, members of Commission 51 endorsed the “Declaration of Activities following the Detection of Extraterrestrial Intelligence” and forwarded it to the IAU council for concurrence. The IAA SETI Committee’s persistent efforts throughout the 1980s on the first “Declaration” was yielding results, with endorsements from the IAA, the Committee on Space Research (COSPAR), members of Commission 51 of the IAU and Commission J of the International Union of Radio Science (URSI). It was noted that once approval was secured from the IAF, the document would be distributed by the IAA to all parties engaged in SETI searches, inviting them to review and consider becoming signatories.

The IAA SETI Committee continued its editorial contributions to special issues of *Acta Astronautica*, focusing on SETI-related topics. During the meeting it was reported that two special issues of *Acta Astronautica* have been published. Jean Heidmann was slated to edit a new special issue of *Acta Astronautica* on SETI, which would feature selected papers from previous IAC’s held between 1987 and 1990 in Brighton, Bangalore, Torremolinos and Dresden.

In 1991 several committee members attended the Third Decennial US-USSR Conference on SETI held in

[‡] In 1982 the Executive Committee of the International Astronomical Union established a commission devoted to the search for extraterrestrial life. Rudolph Pesek and John Billingham assisted the new Commission 51 and Michael Papagiannis was appointed an official representative of the International Astronomical Union to the IAF and accepted membership in the IAA SETI Committee. The IAU Bioastronomy Conferences sponsored by Commission 51 were triennial events until 2007.

Santa Cruz, California, shortly before the dissolution of the Soviet Union.

The 42nd IAC was held in Montreal, Canada in 1991. Six of the eight members present were from the U.S.A. Chair John Billingham reported that the Declaration document was now endorsed by the IAA, IISL, COSPAR, Commission J of URSI and the IAU.

The committee continued its efforts on a draft document addressing the contentious topic of “A Reply from Earth”. It was agreed that a subcommittee, led by Michael Michaud, would be established to refine the existing draft. To reinforce the new subcommittee, a standing task force, headed by John Billingham and consisting of technical and scientific experts, as well as those with expertise in media and societal issues, would be formed. Task force members would be available to assist in interpreting and analyzing any potential SETI signal and in facilitating post-detection communication concerning the discovery.

3.0 Burgeoning SETI Search Support

The year 1992 was an auspicious year for global SETI endeavours. The IAA SETI Committee met and held sessions at the World Space Conference, a special combined meeting of the IAF and COSPAR. This gathering included a Special Symposium on SETI alongside the IAA SETI Sessions and the annual meeting. Dave Black gave the Pesek Lecture on “The Search for Other Planetary Systems: Progress to Date and Future Prospects.”

During the World Space Conference the IAA SETI Committee met on August 30, 1992. The meeting was very well attended with John Billingham (Chair), Ivan Almar (Co-Chair), Jill Tarter (Co-Chair) and several IAA SETI committee members and guests.

On October 12, 1992, shortly after the World Space Congress, NASA launched the High Resolution Microwave Survey. This marked a propitious new age for global SETI efforts and provided a public relations boost for ongoing searches for extraterrestrial signals including the long-running SETI survey at Ohio State University, SERENDIP surveys conducted through the University of Southern California, Berkeley, and SETI programs focusing on the southern sky among others.

3.1 1993 – A Turbulent Year for NASA-funded SETI Activities

Several, if not most, IAA SETI Committee members were actively engaged in various SETI search projects. Securing institutional funding for these projects was

always a challenge. Some SETI search projects were completed by scientists working somewhat surreptitiously outside of their institutional parameters, while others relied on donated equipment and funding.

The initiation of NASA's High Resolution Microwave Survey in 1992 combined with several ongoing SETI radio telescope searches buoyed SETI enthusiasts around the globe as the science of SETI was now validated and endorsed by a staid and distinguished government institution. The IAA SETI Committee had completed the Declaration document and was hard at work on drafting a new document dealing with transmissions from Earth. It was in this upbeat environment that the U.S. Congress suddenly withdrew all funding for anything related to SETI, including the High Resolution Microwave Survey. American SETI researchers were no strangers to controversy and challenges regarding NASA funding for their various activities. Still, the sudden extinguishment of official NASA involvement in the search for extraterrestrial intelligent life was particularly distressing.

The sudden and total withdrawal of NASA funding for all things related to SETI directly impacted several key members of the IAA SETI Committee. The dire news was reflected in the makeup of the attendees at the IAA SETI Committee meeting held during the 44th IAC held in Graz, Austria, where the normally robust NASA SETI contingent was conspicuously absent. Ivan Almar, Co-Chair of the IAA SETI Committee acted as Chair in the absence of John Billingham. The committee expressed deep concern regarding the U.S. Congressional funding decision and explored ways to convey this concern to relevant authorities. Despite the implosion of all NASA-funded SETI activities and its potential repercussions for the small global SETI committee, committee members persevered, continuing to review and discuss ongoing and future SETI initiatives.

A Post-Detection Subcommittee chaired by Govind Swarup received approval from the IAA Board of Trustees. Additionally, a subcommittee on Media and Education was established. The close ties between this subcommittee and the 1991-1992 NASA workshops on the Societal Implications of a Detection were recognized. Progress in this area was anticipated at a SETI and Society Conference planned for Chamonix, France from June 12-18, 1994, organized by John Billingham.

The number of IAA SETI Committee members by early 1994 had increased further to thirty-three[§] as follows:

Billingham, John	U.S.A.
Ivan Almar	Hungary
Jill Tarter	U.S.A.
Francois Biraud	France
David Blair	Australia
Stuart Bowyer	U.S.A.
Peter Boyce	U.S.A.
Aldo Cocca	Argentina
Raoul Colomb	Argentina
Donald Devincenzi	U.S.A.
Carl Devito	U.S.A.
Stephen Doyle	U.S.A.
John Dreher	U.S.A.
Ben Finney	U.S.A.
Samuel Gulkin	U.S.A.
Jean Heidmann	France
William Hilton	United Kingdom
Hisahi Hirabayashi	Japan
Nandasiri Jasentuliyana	Sri Lanka
Jan Jugaku	Japan
Nikolai Kardashev	C.I.S.
Michael Klein	U.S.A.
Vladimir Kopal	C.S.F.R.
Vladimir Kotelnikov	C.I.S.
Michael Michaud	U.S.A.
Edward Olsen	U.S.A.
Jorg Pflaederer	Austria
Wilber Pritchard	U.S.A.
John Rather	U.S.A.
V.I. Slysh	C.I.S.
Vladimir Strelitnikij	C.I.S.
Mieczyslaw Subotowicz	Poland
Govind Swarup	India
Donald Tarter	U.S.A.

At the 45th IAC in Jerusalem, Israel in 1994, Nicolai Kardashev presented the Rudolph Pesek Lecture "Space Programmes for SETI".

The growing evidence for extrasolar planets was of great interest to the IAA SETI Committee, the broader scientific community and the general public. An estimate of the number of stars with planetary systems, including those with planets similar to Earth, would inform the Drake equation calculation in estimating the number of extraterrestrial intelligent species enabling

[§] The membership on the IAA SETI Committee was and is remains somewhat fluid due to limited terms and rotation on and off the committee. In addition, the administration of the committee requires adherence to the broader IAA rules.

communication and technology that could be recognized by humans.

Rather surprisingly, the response in endorsements from SETI researchers regarding Declaration document was lukewarm, with only three endorsements received to date. There was a reluctance on the part of institutions and individuals to sign and commit to the voluntary principles outlined in the Declaration.

Work on the IAA SETI Committee's White Paper on "A Decision Process for Examining the Possibility of Sending Communications to Extraterrestrials Civilizations" continued. John Billingham, Michael Michaud and others worked diligently with the IAA to streamline the document, now referred to as "Position Paper", and seek endorsement from the IAA Board of Trustees. This was duly completed and George Mueller, President of the IAA, forwarded the Position Paper, along with a letter from John Billingham, to the Board of Directors of IISL for their approval.

New members proposed for the IAA SETI Committee were Michael Ashkenazi (Israel), Roberta Vaile (Australia), Claudio Maccone (Italy), Kelvin Wellington (Australia), Roger Malina (U.S.A.), Christiano Cosmovici (Italy), and Tom Pierson, (U.S.A.). Jean Heidmann continued work on collecting papers for publication in *Acta Astronautica*, intending to produce a third special issue.

4.0 Internet, Email, Exoplanets, and Martian Fossils

By the mid-1990s, committee members took advantage of and grappled with the advent of new email networks and the implications of the rapid growth of instant communication media. Jean Heidmann attempted to build a SETI E-mail distribution network to facilitate communication among SETI researchers, especially concerning candidate signals and other topics of interest. However, a small survey revealed that some respondents would be hesitant to immediately divulge details of their discovery and were concerned about security of data on an email network.

Public interest in SETI continued with the founding of SETIQuest Magazine, founding of the SETI League, a group of radio astronomy and other amateur and professional enthusiasts, continued support from the Planetary Society for SETI projects, and other initiatives.

In 1995 during the 46th IAC in Oslo, Norway the Lunar SETI Studies Subcommittee was established. Its purpose was to examine aspects of dedicating a radio

observatory on the far side of the moon to ensure the long-term future of SETI. IAA SETI Committee member Jean Heidmann had long been involved in efforts to protect areas on the lunar farside from radio interference.

Additionally, 1995 brought in a groundbreaking discovery – the first confirmed exoplanet orbiting a main sequence star, 51 Pegasi. This momentous finding, along with subsequent identification of exoplanets using various observation methods, bolstered the prospects of discovering signs of extraterrestrial life.

Also in 1995, a group of SETI researchers directly affected by the cancellation of the NASA SETI program in 1993 started observations on Project Phoenix. Herculean efforts by Jill Tarter, John Billingham, Barney Oliver and others sourced private donations administered by the SETI Institute in California to continue the search.

The surprise announcement in 1995 of potential fossils found in a Martian meteorite, and the subsequent controversy was observed carefully by the IAA SETI Committee. Conclusive evidence and, of note, close tracking of who has access to the information before a global announcement are two factors that impacted the somewhat chaotic events leading up to a news conference and release of a statement from the White House. The events regarding the Martian rock are described vividly by Dick and Strick [6].

The events surrounding the announcement of the potential Martian fossil brought heightened attention to the IAA SETI Committee's Declaration document and ongoing work on potential transmissions from Earth..

The IAA provided full support for the IAA SETI Committee's initiatives, particularly in introducing what was now an IAA Position Paper proposing a decision process for examining the possibility of sending communications to extraterrestrial civilizations. In 1996, George Mueller, President of the IAA, sent letters to over sixty United Nations Committee on the Peaceful Uses of Outer Space (UN COPOUS) member countries, requesting them to forward the letter and attached documents to the Chair of UN COPOUS for discussion and potential action. The hope was that one or more countries would introduce it for discussion within UN COPOUS.

The 47th annual IAC in 1996 took place in Beijing, China. Beijing had originally been selected as the site for the 1989 40th IAC, but due to the Tiananmen Square protests and martial law declaration by Chinese

authorities the location for the IAC was changed to Torremolinos, Spain.

The Beijing Congress in 1996 featured SETI papers on the prevalence of earth-like planets, SETI observations from the moon, design concepts for one km sq radio telescopes, including Chinese proposals for a similar facility. Additionally, SETI facilities in Italy at the Medicina/Noto sites were described.

The IAA SETI Committee held its annual meeting on October 5, 1996. Nine members of the committee (Kelvin Wellington, Ernst Fasan, Donald Tarter, Wm. Hilton, Stephen Doyle, Aldo Cocca, Claudio Maccone, Roger Malina and Vladimir Kopal) were present in Beijing, along with John Billingham, Chairman and Jill Tarter, Co-Chair and Acting Secretary. Guests included several future members of the IAA SETI Committee including Stelio Montebugnoli, Albert Harrison, Robert DeBiase, and Allen Tough. During the meeting it was announced that a Scientific-Legal Roundtable on SETI and Society would be scheduled during next year's IAC. Although Jean Heidmann was not present in Beijing, it was noted that he prepared a paper entitled "SETI from the Moon: An Invitation to COSPAR" describing his activities as Chair of the Lunar SETI Study Subcommittee of the IAA SETI Committee. He continued advocating for participation by other SETI Committee members in the programs of COSPAR Section F on Exobiology.

New members proposed and accepted for IAA SETI Committee membership were Allen Tough, Canada, Stuart Bowyer, U.S.A., Rod Davies, U.K., Shuhua Ye, People's Republic of China, Rendong Nan, People's Republic of China, Peter Sturrock, U.S.A., Martin Rees, U.K., Ragbir Bhathal, Australia, Dan Werthimer, U.S.A., Carol Oliver, Australia, and David Whitehouse, U.K. It was noted that there are no committee members from Germany, despite that country's strong radio astronomy sector.

Govind Swarup successfully established a standing SETI Post-Detection Subcommittee on Science and Technology. Kelvin Wellington, Jill Tarter, Claudio Maccone, Aldo Cocca and Robert DeBiase agreed to participate in the subcommittee dedicated to Lunar SETI studies.

Jill Tarter provided an update on the IAU Bioastronomy meeting in Capri and also reported on the compilation of an archive containing valuable information on various SETI projects and observations. Jean Heidmann discussed ongoing efforts to secure support for a study aimed at proposing the deployment of a small research antenna system within the Saha

crater on the far side of the moon. This initiative would significantly enhance SETI observations.

By the time of the IAA SETI meeting in Beijing, adoption of the World Wide Web was increasing exponentially around the globe. An ALLSETI newsgroup established by Roberta Vaile and managed by Carol Oliver was in operation. Stelio Montebugnoli reported on the construction of a new 70 m radio antenna in Sardinia and its possible utilization for SETI experiments.

5.0 1997 – John Billingham Steps Down

The IAA SETI Committee annual meeting at the 48th IAC in Turin, Italy in 1997 was well attended with 21 members and five guests. Of note was the presence of four current and future Chairs of the committee; John Billingham, Jill Tarter, Seth Shostak, and Claudio Maccone. Chair John Billingham opened the meeting with administrative matters, including his resignation as the SETI Committee Chair and subsequent endorsement of the IAA Board of Trustees for Jill Tarter as the new Chair. According to the IAA, the SETI Committee at this time consisted of 46 members of which 32% were from the United States, 11% from Australia, and 9% from each of the United Kingdom, Russia and Italy. Other members were from Hungary, France, Israel, Austria, Japan, Sri Lanka, Czech Republic, Argentina, Poland, and India.

The IAA SETI Committee continued its administrative work in organizing the SETI Review Sessions at the annual IAC and cooperating with other global SETI efforts and meetings. Jean Heidmann diligently devoted time and effort to editing special issues of *Acta Astronautica*. In addition, he was asked by the IAA to lead a study and prepare a report regarding protecting and establishing a radio observatory on the farside of the moon. Work continued regarding the IAA Position Paper concerning a possible reply to a detected signal.

John Billingham reported that subcommittees under the SETI Committee had been established, and their formal approval by the Board of Trustees of the IAA was requested. The four subcommittees were as follows:

- Issues of Policy Concerning Communications with Extraterrestrial Intelligence (Chair – Michael Michaud, U.S.A.)
- SETI Post-Detection Science and Technology (Chair – Govind Swarup, India)
- Lunar SETI Studies (Chair – Jean Heidmann, France)

- Media and Education (Co-Chairs – Roger Malina (U.S.A.) and Stephen Doyle (U.S.A.)

The Subcommittee on Issues of Policy Concerning Communications with Extraterrestrial Intelligence convened in Turin at the IAC. Michael Michaud, Chair, reported on a vexing issue facing the subcommittee; the long period of time it was taking for radio astronomers, particularly observatory directors, to agree to adhere to the “Declaration of Principles Concerning Activities Following the Detection of Extraterrestrial Intelligence”. He also reported that of the 63 Foreign Affairs Ministers contacted regarding introducing the issue to the United Nations, seven responded and none would agree to introduce the subject, although the Government of Australia stated that it would support another government’s initiative. In light of this tepid response, it was recommended that the President of the IAA make a presentation to the UN COPOUS or one of its subcommittees and that presentation would become a United Nations document as an annex to the subcommittee’s report. Michael Michaud was to follow up with the United Nations Office of Outer Space Affairs Director Nandasiri Jasentuliyana and with the Chair of the IAA SETI Committee. Members of the Subcommittee reported on their individual endeavours on messages from Earth, the topic of sending a reply and what it would be, but there was little enthusiasm among members to collectively work on a “message from Earth”. The term “active SETI” started to be used to describe the transmission of messages from Earth.

The resignation of John Billingham as Chair marked the end of an era where proactive and tireless promoters of the importance of SETI research, Rudolph Pesek and John Billingham, advocated not only for IAA SETI Committee activities, but for all SETI initiatives globally. Both worked hard to ensure the inclusion of the societal ramifications of the detection of extraterrestrial intelligent life in the universe into IAA SETI Committee deliberations. Jill Tarter, the new Chair, was heavily involved with these initiatives and would carry the work forward.

6.0 Presenting at the United Nations Committee on the Peaceful Uses of Outer Space

In 1998, the IAF held its 49th annual IAC in Melbourne, Australia. Jill Tarter chaired the annual IAA SETI committee meeting, which was also attended by Co-Chair Ivan Almar and, notably, by Jean Michel Contant, Secretary General of the IAA. Jean Michel Contant reported on the status of the IAA Position Paper with UN COPUOS and his presentation made to UN COPUOS in Vienna, Austria, June 3, 1998. During this presentation, a request was made to UN COPUOS to

consider the possibility of a subsequent presentation to the Technical Subcommittee of UN COPUOS on the question of transmitting a reply from Earth.

The Melbourne SETI Committee meeting included reports of SETI initiatives in the U.S.A., India, Australia, and Italy. Of interest was the inauguration of a virtual Astrobiology Institute by NASA. The four subcommittees established by the IAA SETI Committee were active and met during the IAC. Claudio Maccone reported on a meeting chaired by committee member Jean Heidmann on “Protection of Part of a Celestial Body for the Scientific Benefit of Humankind: the Lunar Farside Crater Saha Proposal” and held in Nagoya, Japan. The focus of protection was shifting from the previous Saha crater to other areas on the dark side of the moon.

7.0 A New Century and A New Era Begins for SETI

The 1990s drew to a close, with the IAA maintaining its robust collaboration with the IAU’s Commission 51. During this period, the IAA assisted in planning for four IAU Bioastronomy conferences at various locations: Val Cenis, France (1990), Santa Cruz, U.S.A. (1993), Capri Italy (1996) and finally in Hawaii, U.S.A. in 1999. A new multidisciplinary science termed astrobiology** was emerging from NASA, reflecting extrasolar planet discoveries and the enhanced interest in the possibility of life beyond Earth.

In October, 1999, the International Astronautical Federation (IAF) held its 50th IAC in Amsterdam, Netherlands. John Billingham presented the Pesek Lecture during the SETI review session and proposed a systematic approach to addressing the detection of extraterrestrial intelligent life, whether through serendipitous discovery, active searches, or responses to transmissions from Earth. This session featured technical discussions on Project Phoenix and the Southern SERENDIP project, as well as broader talks examining the implications of a verified detection. This

** The terms astrobiology, bioastronomy, exobiology at this time were particularly described. The bioastronomy conferences held by the IAU included talks on the search for intelligent extraterrestrial life. By the end of the 1990s, NASA defined astrobiology as the study of life in the universe using the techniques and facilities of the space program to answer basic questions including 1) How does life begin and evolve (2) Is there life elsewhere in the universe (3) What is the future of life on Earth and beyond. Specifically excluded was SETI-related research activities to search for signs of extraterrestrial intelligence or technology.

issue gained prominence following reports of a potential signal from the star EQ Pegasi, which quickly garnered media attention and examination of the IAA Declaration document.

The final IAA SETI Committee meeting of the 1990s convened with Chair Jill Tarter, Co-Chairs Ivan Almar and Tom Pierson, and Secretary Jean Heidmann in attendance. Members present included John Billingham, Robert DeBiase, Stephen Doyle, Ernst Fasan, Albert Harrison, Vladimir Kopal, Roger Malina, Claudio Maccone, Michael Michaud, Stelio Montebugnoli, Ramirez de Arellao, Salvatore Santoli, Seth Shostak, Patricia Sterns, Les Tennen, Allen Tough, and Doug Vakoch. The numbers of members of the IAA SETI Committee at this time numbered about 50, with each member serving a three-year term. Members were from the U.S.A., Hungary, Australia, Argentina, Italy, United Kingdom, Austria, France, Russia, Czech Republic, Mexico, China, India, and Canada. Various SETI projects were reviewed at the meeting including the spectacularly successful SETI@home project hosted at the University of California at Berkeley, now with 1.2 million subscribers, a SETI 2020 vision document prepared by the SETI Institute, optical SETI searches, SETI search efforts in Italy, Australia, and Argentina.

The committee grappled with the same contentious issues at the end of the decade as ten years earlier; the issue of how to respond to a verified detection of extraterrestrial intelligence, and the emerging question of whether to actively transmit messages from Earth.

Following the successful endorsement of the Declaration document by various bodies, the IAA SETI Committee focused on raising awareness of the Declaration and introducing the Position Paper document to the UN COPOUS. The IAA continued to advocate for inclusion of this topic. Thanks to the dedicated efforts of several IAA SETI Committee members and their collaborators, Jill Tarter, with the support of several IAA SETI Committee members and their collaborators, Jill Tarter, with the support of and introduction by the IAA's Jean Michel Contant, presented the matter to UN COPOUS in June, 2000.^{††} This presentation marked a significant milestone in the tireless efforts of individuals like John Billingham, Jill

^{††} The presentation asked for inclusion of the IAA Position Paper in the Report of UN COPOUS and that a copy of the IAA Position Paper, including its appendices, be submitted to the UN Office for Outer Space Affairs for information and possible further use by the Secretary General or by, UN COPOUS or other relevant bodies of the United Nations.

Tarter, Michael Michaud, Ernst Fasan, and their colleagues within the IISL to bring this issue to the attention of the United Nations for information and consideration.

8.0 Conclusion

The IAA SETI Committee remains an vital permanent study group under the umbrella of the International Academy of Astronautics. Rudolph Pesek first introduced the concept of SETI to his IAA colleagues in 1965, and the IAA supported its first SETI symposium in 1972 at the 23rd IAC held in Vienna, Austria. The IAA SETI Committee was subsequently established in 1974. Its mandate and general structure, supported by the IAA, remained consistent throughout the 1990s and continues to this day.

To comprehensively approach and conduct a scientific search for extraterrestrial intelligent life in the universe, one must not only grasp the science and technology involved, but also consider the cultural, philosophical, and societal implications of discovering that humanity is not alone. The IAA SETI Committee consistently and diligently explores all aspects of the quest for evidence of extraterrestrial intelligent life, and it strives to understand and prepare for all potential consequences to humanity in the event of a verified detection. Committee members and their collaborators provide a strong, cross-disciplinary foundation for examining all facets of the search for extraterrestrial intelligent life in the universe. The Declaration document completed by the IAA SETI Committee in 1989 and its subsequent revisions remain consequential as does the committees continued diligent work, with support from the IAA, on the issue of sending deliberate transmissions from humanity to space.

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